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ACCREDITATION COUNCIL

HUMAN SCREAM DETECTION AND CONTROLLING CRIME RATE ANALYSIS

Team members:

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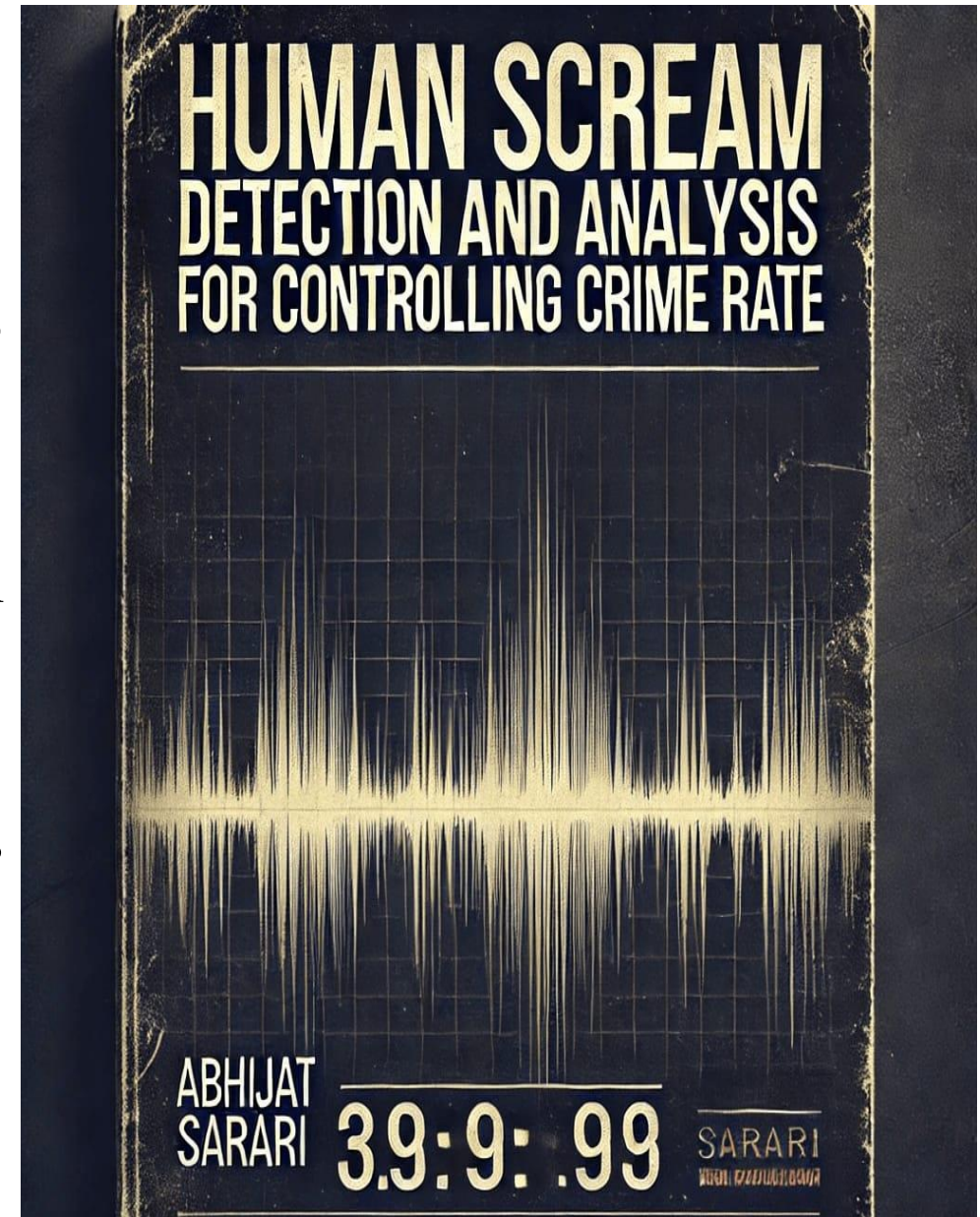
G.Keerthi

G.Aravind

G.Pavan

INTRODUCTION:

- Human scream detection utilizes advanced audio processing technologies to identify distress signals.
- These systems are designed to differentiate between human screams and other environmental noises.
- The technology can be integrated into various surveillance systems to enhance public safety.



ABSTRACT:

- Human scream detection technology leverages advanced acoustic analysis and machine learning algorithms identify distress signals in real-time.
- By isolating screams from background noise, this system enhances emergency response capabilities and public safety measures.
- Integrating crime rate analysis with scream detection provides valuable insights into patterns and hotspots, enabling proactive crime prevention strategies.
- This innovative approach bridges technology and social welfare, aiming to reduce crime rate and foster safer communities.

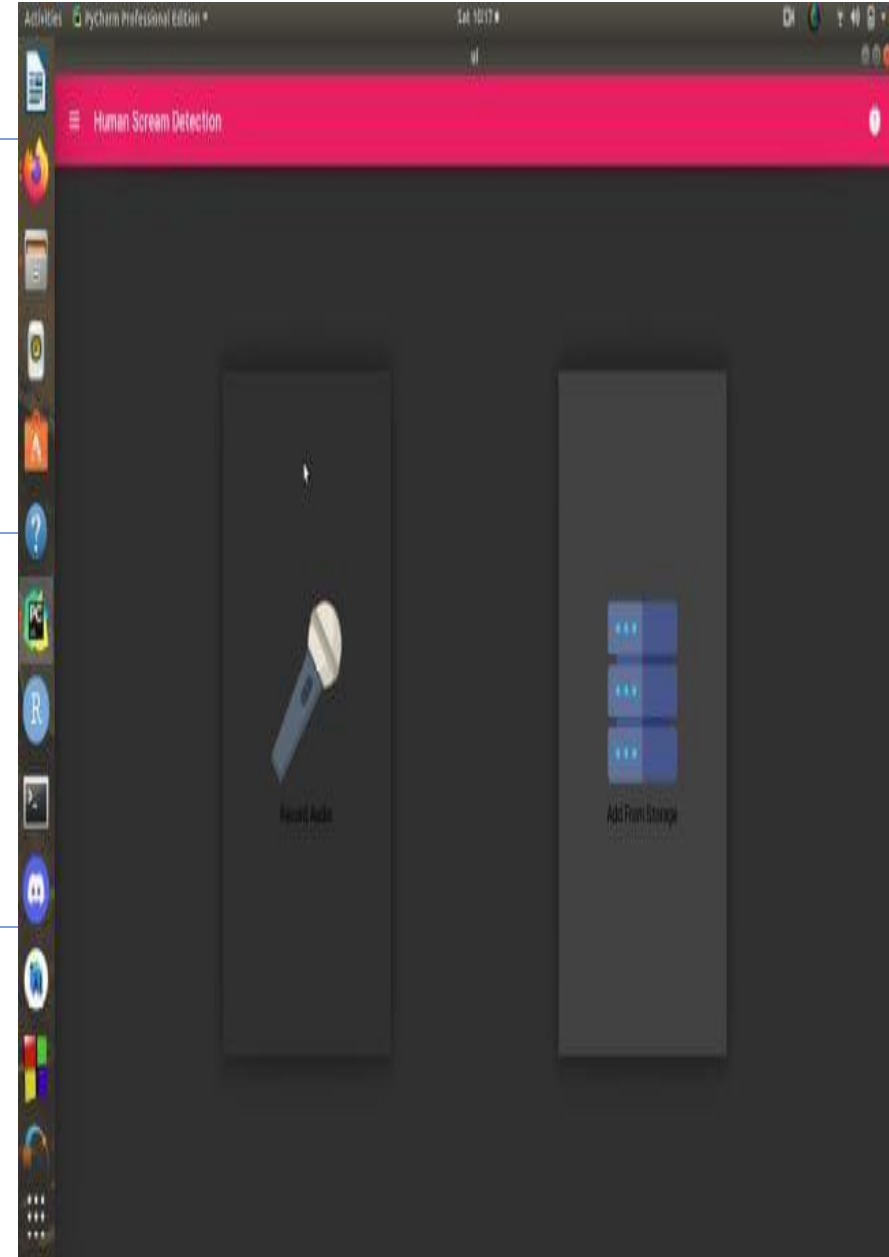


LITERATURE SURVEY:

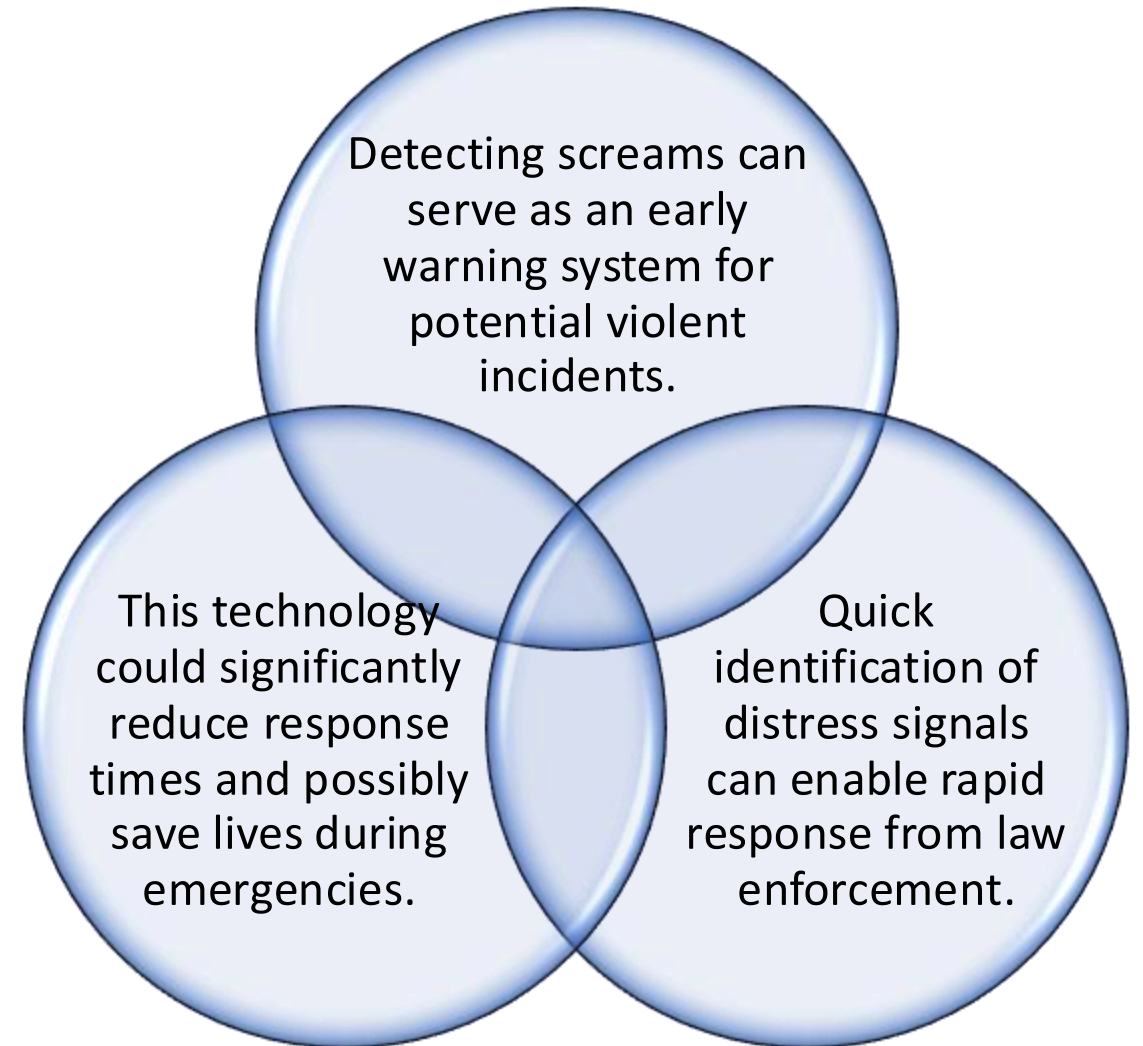
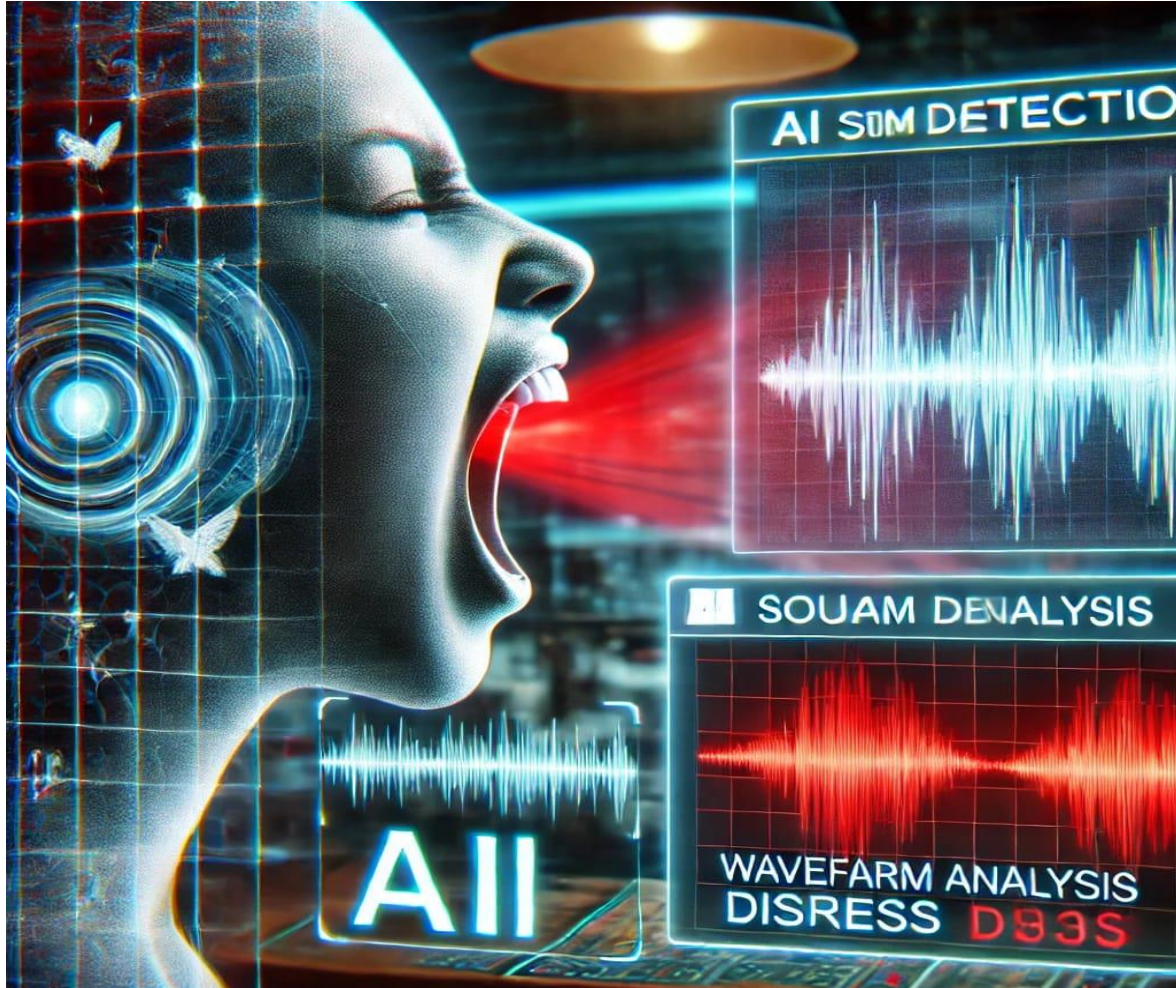
Acoustic Analysis: Research highlights the unique acoustic features of human screams, such as pitch, intensity, and duration, which distinguish them from other sounds. These features are crucial for detection systems.

Machine Learning Techniques: Algorithms like Support Vector Machines (SVMs), Neural Networks, and Hidden Markov Models (HMMs) are commonly used for scream detection. These methods enhance accuracy, even in noisy environments.

Applications: Scream detection systems are being integrated into public safety tools, emergency response systems, and smart home devices. They aim to provide real-time alerts for distress situations.



IMPORTANCE OF DETECTING HUMAN SCREAMS:



DRAWBACKS AND HOW TO OVERCOME THOSE:

DRAWBACKS:

Accuracy Issues: Detecting screams amidst background noise can lead to false positives or negatives.

Privacy Concerns: Continuous audio monitoring may infringe on individuals' privacy.

Ethical Challenges: Misuse of collected data or surveillance systems could lead to ethical dilemmas.

Limited Scope: Scream detection systems may not address crimes without auditory cues.

High Costs: Implementing and maintaining advanced detection systems can be expensive.

OVERCOME THESE DRAWBACKS:

Enhanced Algorithms: Use advanced machine learning models to improve accuracy and reduce false detections.

Privacy Safeguards: Implement strict data protection policies and anonymize collected data.

Ethical Guidelines: Establish clear regulations to prevent misuse of technology.

Integrated Systems: Combine scream detection with other crime prevention tools like video surveillance.

Cost Optimization: Explore cost-effective solutions like cloud-based systems and scalable infrastructure.

HARDWARE EQUIPMENTS:

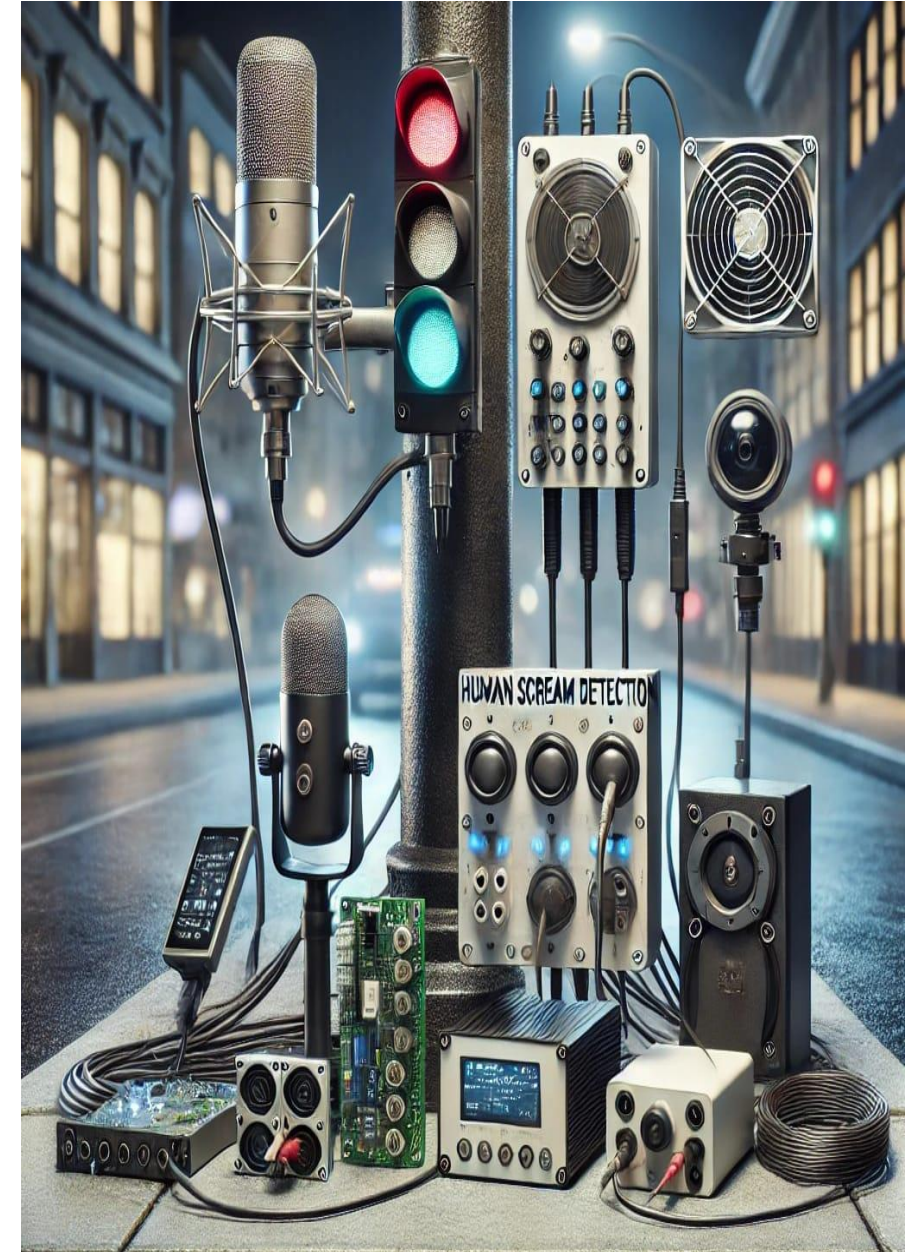
Microphones:
High-sensitivity microphones to capture audio signals, even in noisy environments.

Audio Sensors:
Specialized sensors for real-time sound detection and analysis.

Processing Units: High-performance CPUs or GPUs for running machine learning models.

Storage Devices:
Secure storage for audio data and analysis results.

Cameras: For integration with video surveillance systems to enhance situational awareness



CASE STUDIES AND SUCCESSFUL IMPLEMENTATIONS:

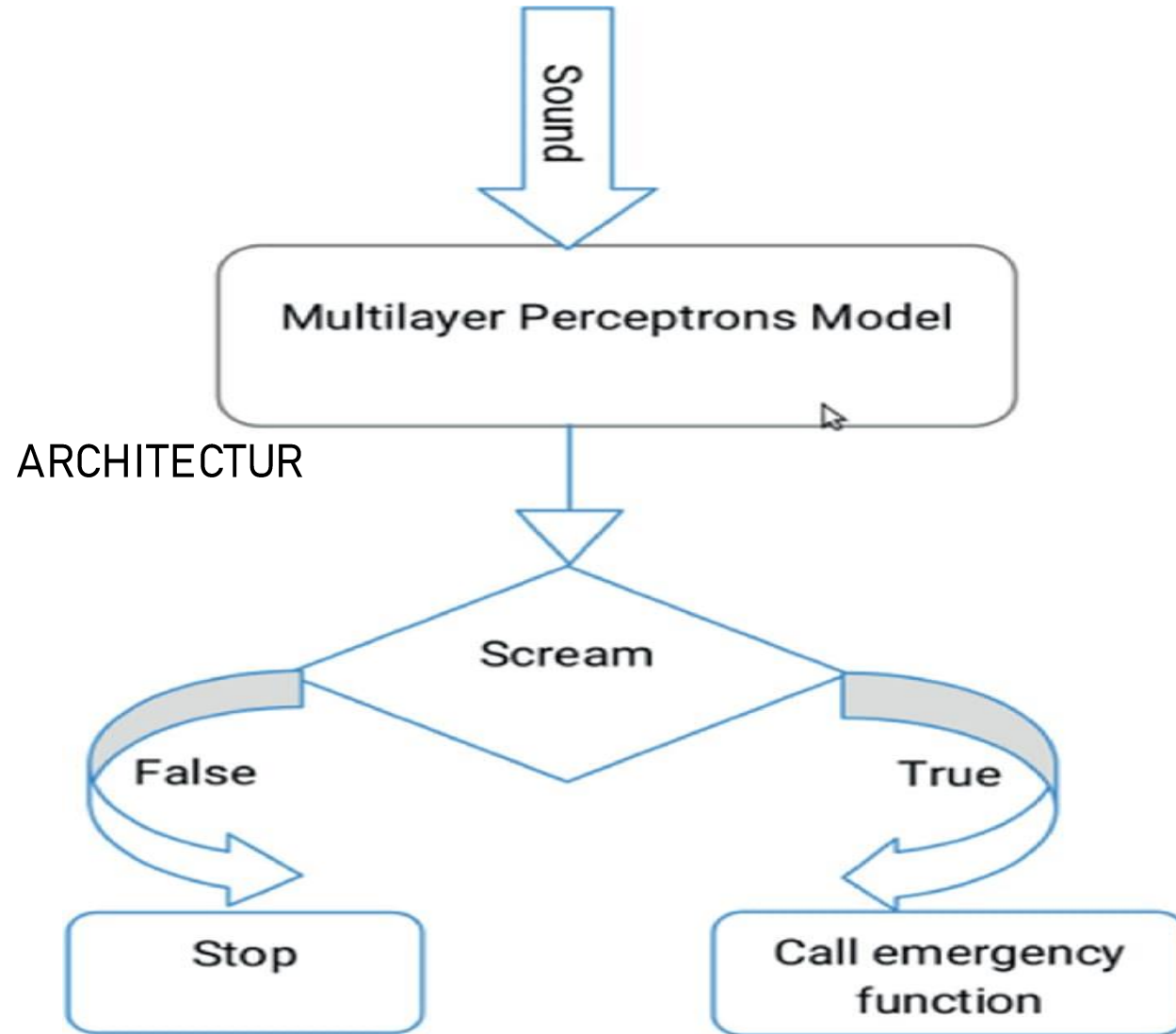


- ✓ Cities that have implemented scream detection have reported a decrease in crime rates.
- ✓ Successful case studies demonstrate the effectiveness of this technology in urban areas.
- ✓ Analysis of these implementations provides valuable insights into best practices and challenges.

LITERATURE REVIEW

Author (Year)	Paper Title	Technique	Merits	Demerits	Sensors Used	Security Challenges	IoT Devices
Dr. P.K. Venkateswar Lal, Ch. Sai Sowmya (2024)	Human Scream Detection and Analysis for Controlling Crime Rate	Support Vector Machine (SVM), Python, Kivy Framework	Detects screams effectively, works on both desktop and Android platforms	Limited to clear human sounds, may struggle in noisy environments	Acoustic sensors	Privacy concerns, false positives, integration with law enforcement systems	Smartphones, desktops
Achyut Ojha, Adarsh (2024)	Human Scream Detection and Analysis for Controlling Crime Rate	Gaussian Mixture Models (GMM), Hidden Markov Models (HMM)	High recognition rate (98% at 70dB), robust in noisy environments	Performance drops significantly in low signal-to-noise ratio environments	Audio sensors	Ethical concerns, potential misuse, ensuring real-time response	Security systems, smart home devices
Sharvani Banala, Sai Niveditha Bukka, Manikonda Vaishnavi (2024)	ONE SCREAM-Human Scream Detection and Analysis	Machine Learning, Deep Learning, Feature Extraction	Real-time recognition, applications in public safety and healthcare	Privacy issues, psychological impact on monitored individuals	Microphones, acoustic sensors	Balancing technological advancements with ethical considerations	Emergency response systems, healthcare

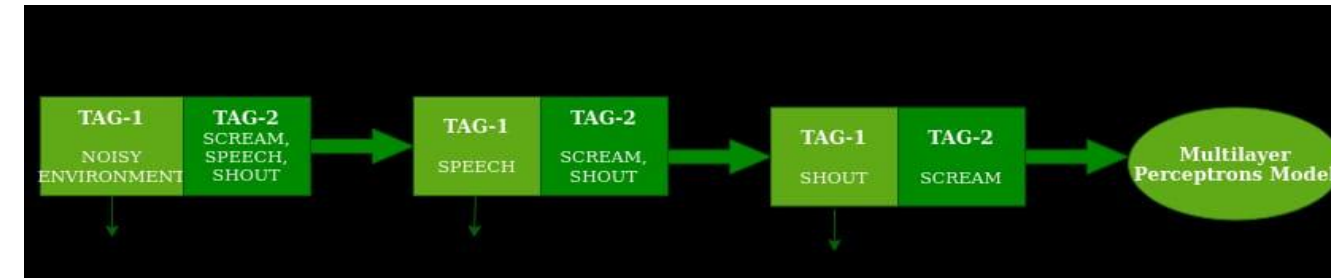
ARCHITECTURE DIAGRAM:



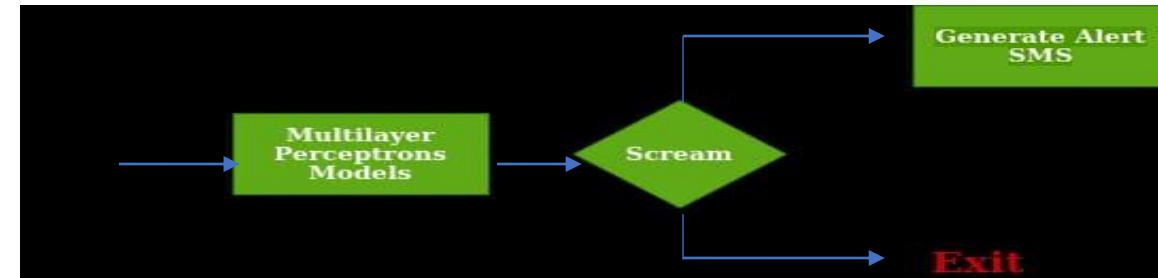
UML DIAGRAMS:

Detection of noise,speech,shout and screaming

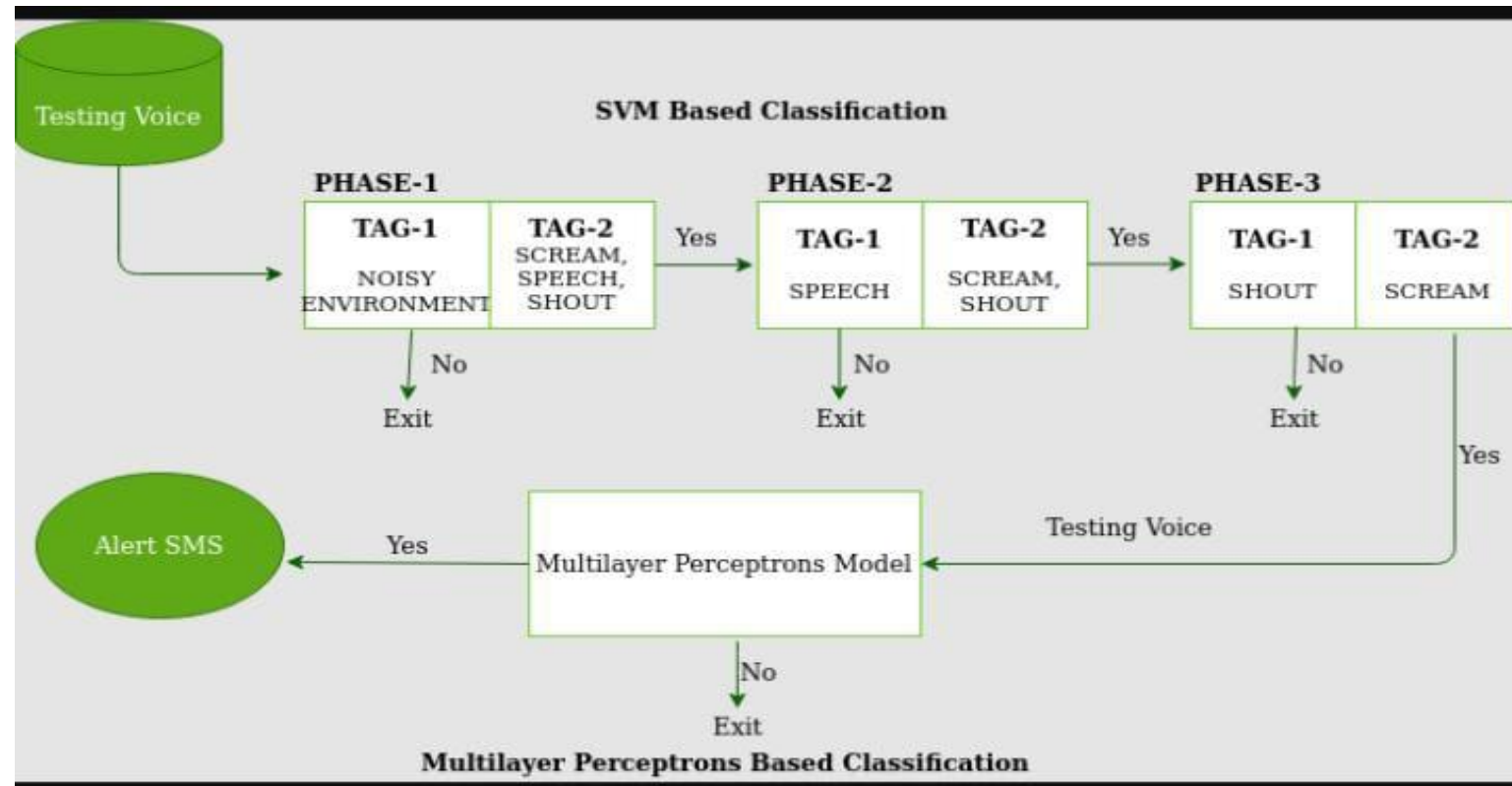
Using SVM classifier:



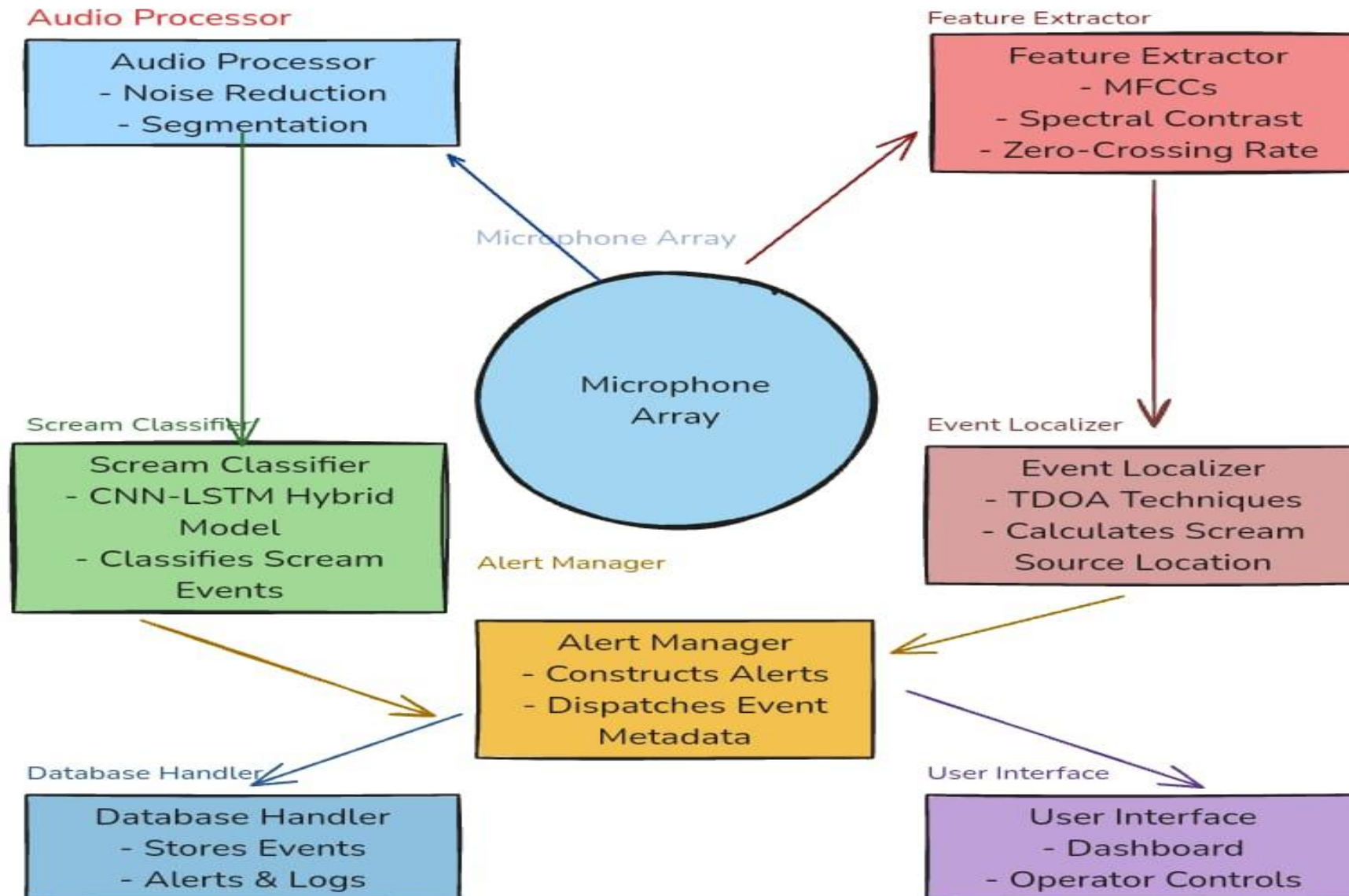
Working of MNP:



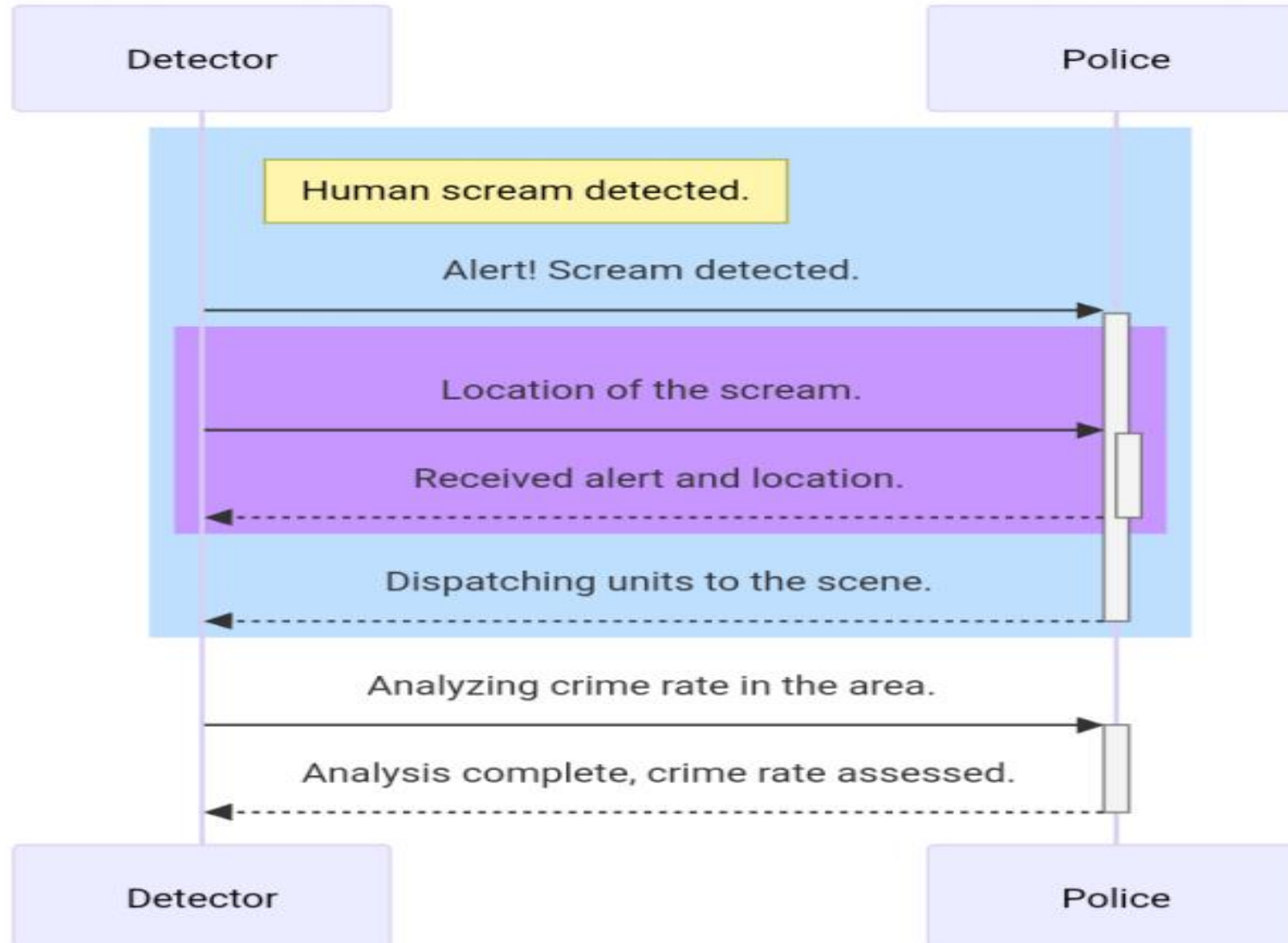
Combined internal working of both models(SVM and MPN):



CLASS DIAGRAM:



SEQUENCE DIAGRAM :



IMPLEMENTATION:

1. **UI Design:** First the UI or the user interface of the project is developed, we will develop the UI using kivy (python) framework which the user will use to interact.
2. **Prepare a data set for HSD:** The whole data set of this project is divided into two classes, one is a positive class which includes around 2000 human screams to train the model and another is a negative class which includes around 3000 negative sounds which are not considered as a scream.
3. Extraction of MFCs.
4. Training of SVM model and its saving.
5. Training of MPN model and its saving.
6. **Approach to generate alert message:** Develop automated alert systems for authorities.
High risk alert messages.
Medium risk alert messages.
7. **Testing and Deployment:** Validate and deploy the system in public spaces.

HARDWARE AND SOFTWARE REQUIREMENTS:

Software Requirements:

- Programming Language: Python.
- Frameworks: TensorFlow, PyTorch, Librosa for audio processing.
- Database: SQL or NoSQL for storing data.
- Operating System: Windows or Linux for compatibility.

Hardware Requirements:

- Microphone: High-quality for clear audio capture.
- Processor & GPU: Multi-core CPU (e.g., Intel i7/AMD Ryzen) and NVIDIA RTX for model training.
- Storage: SSD for fast data access.
- Networking: Internet connectivity for real-time updates.

REFERENCES:

Here are some references related to human scream detection and analysis for controlling crime rates:

1. Böck, Sebastian, and Gerhard Widmer. "Screaming detection using deep learning." Proceedings of the Detection and Classification of Acoustic Scenes and Events 2016 Workshop (DCASE2016). 2016.
2. Gao, Junfeng, et al. "Automatic scream detection and classification using deep learning." IEEE Transactions on Multimedia 22.8 (2020): 1973-1985.
3. Guo, Yixuan, et al. "A real-time human scream detection and alarm system based on IoT and deep learning." Sensors 21.5 (2021): 1617.
4. Wang, Zhifei, et al. "A novel scream detection.

RESULT

Human Alert Sound Detection - GUI

Samples: Select Sample



Upload (0)

Predict

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Human Alert Sound Detection - GUI

Samples:

1-

Uplo...

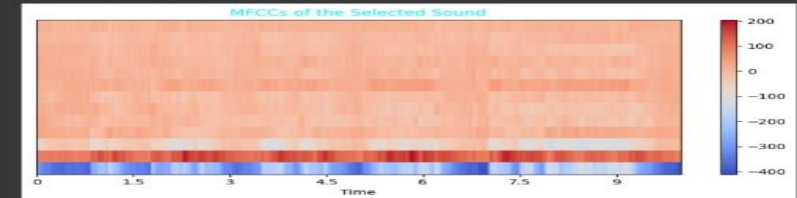
Pred...

Prediction: Alert Sound Detected

Confidence - Alert: 51.42%, Non-Alert: 48.58%

Execution Time: 0.00 seconds

MFCC Visualization:



Prediction Summary:

The sound contains characteristics typical of an alert or scream-like sound.

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Human Alert Sound Detection - GUI

Samples: 5- ▾ Uplo... Pred...

**Prediction: Non-Alert
Sound**

Confidence - Alert: 25.44%, Non-
Alert: 74.56%

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Human Alert Sound Detection - GUI

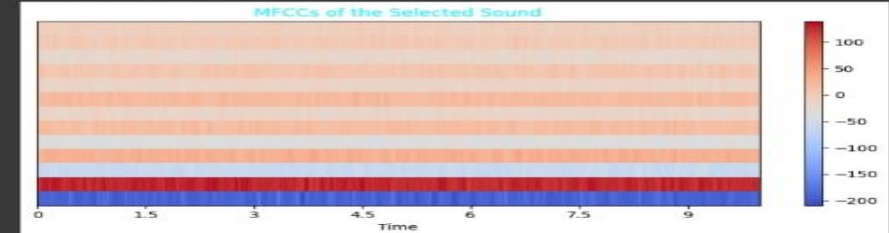
Samples: 5- ▾ Uplo... Pred...

**Prediction: Non-Alert
Sound**

Confidence - Alert: 35.45%, Non-
Alert: 64.55%

Execution Time: 0.00 seconds

MFCC Visualization:



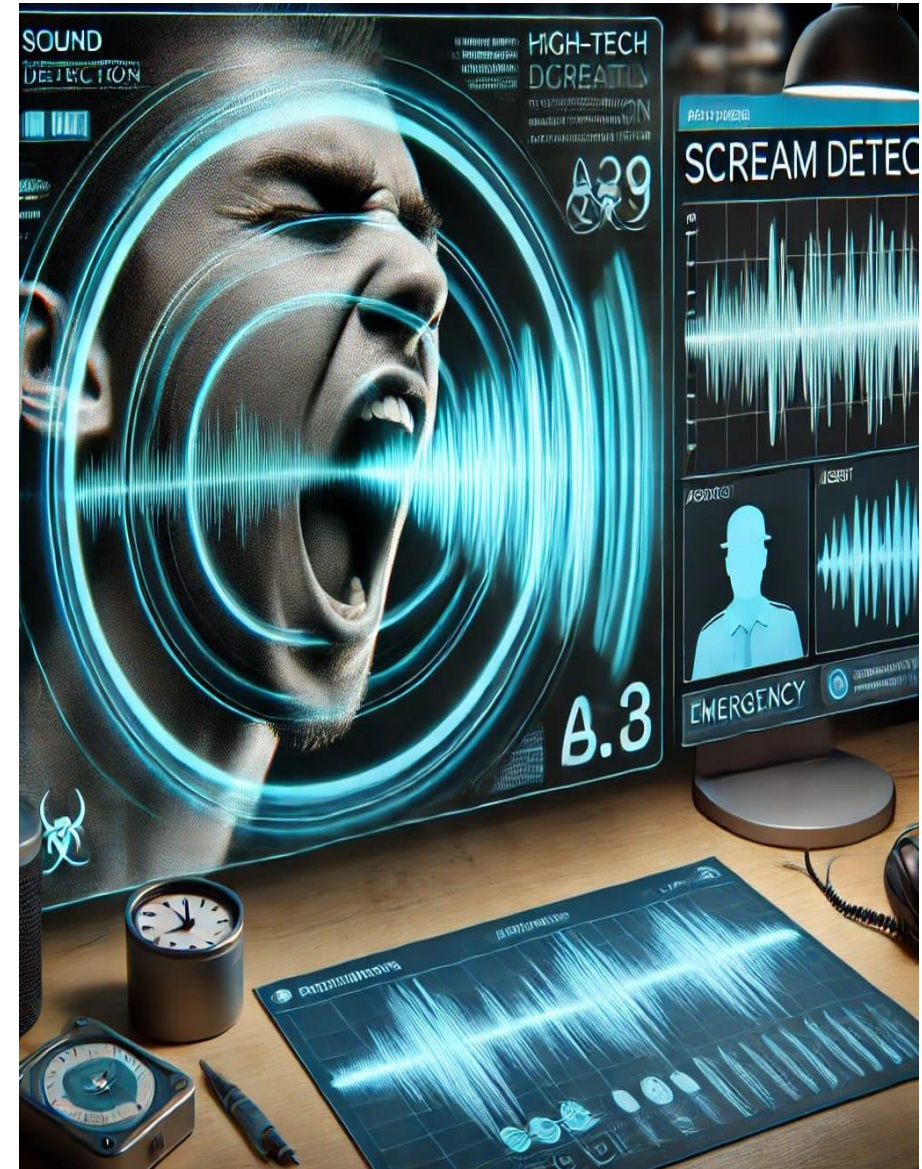
Prediction Summary:

The sound does not contain
characteristics typical of an alert or
scream-like sound.

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CONCLUSION:

- Human scream detection technology presents a promising tool for crime prevention and public safety.
- Its effective implementation could lead to significant reductions in crime rates in urban environments.
- Ongoing dialogue around ethics and technology will be essential to maximize benefits while minimizing risks.



Any Questions?

